🔑 **Answer Key — Teacher Copy · Lesson 5-4**

MSTAR PRACTICE WORKSHEET

Lesson 5-4: Apply Area Concepts to Solve Problems

Grade 6 Mathematics · Standard 6.GR.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

An L-shaped room is made of two rectangles: 10 ft × 6 ft and 4 ft × 3 ft. What is the total area?

A. 12 sq ft

B. 60 sq ft

C. 72 sq ft

D. 72 ft

Area 1 = 10 × 6 = 60 sq ft. Area 2 = 4 × 3 = 12 sq ft. Total = 60 + 12 = 72 sq ft.

Question 2 SELECTED RESPONSE — CORRECT: B

A rectangular patio is 15 ft × 10 ft with a 5 ft × 4 ft rectangular flower bed cut out. What is the remaining area?

A. 20 sq ft

B. 130 sq ft

C. 150 sq ft

D. 170 sq ft

Patio = 15 × 10 = 150 sq ft. Cutout = 5 × 4 = 20 sq ft. Remaining = 150 − 20 = 130 sq ft.

Question 3 SELECTED RESPONSE — CORRECT: D

Find the area of an L-shape made of a 5×3 rectangle and a 2×4 rectangle.

A 8 sq units

B 15 sq units

C 20 sq units

D 23 sq units

5 × 3 = 15. 2 × 4 = 8. Total = 15 + 8 = 23 sq units.

Question 4 SELECTED RESPONSE — CORRECT: D

When should you subtract areas instead of adding them to find the area of a composite figure?

A When the shapes overlap

B When one shape is larger than the other

C Always subtract — never add

D When a piece is cut out or removed from a larger shape

Subtract when a piece is removed from a larger shape (like a window cut from a wall). Add when two shapes are joined together (like an L-shape).

Question 5 SELECTED RESPONSE — CORRECT: C

A 7 ft by 5 ft rectangle has a triangle (base 7 ft, height 4 ft) on top. Add the two areas. What is the total area?

A 14 sq ft

B 35 sq ft

C 49 sq ft

D 63 sq ft

Rectangle = 7 x 5 = 35 sq ft. Triangle = 1/2 x 7 x 4 = 14 sq ft. The pieces are joined, so add: 35 + 14 = 49 sq ft.

Question 6 SELECTED RESPONSE — CORRECT: D

Plan 2 — L-shaped lobby: decompose the L into two rectangles, one 9 ft by 6 ft and one 7 ft by 3 ft, then ADD. What is the total floor area?

A. 21 sq ft

B. 54 sq ft

C. 63 sq ft

D. 75 sq ft

First rectangle = 9 x 6 = 54 sq ft. Second rectangle = 7 x 3 = 21 sq ft. Joined L-shape, so add: 54 + 21 = 75 sq ft.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

An L-shaped room splits into two rectangles: one is 12 feet by 8 feet, the other is 6 feet by 5 feet. What is the total floor area?

A. 30 square feet

B. 62 square feet

C. 96 square feet

D. 126 square feet

12 × 8 = 96 and 6 × 5 = 30. The total is 96 + 30 = 126 square feet.

- **A:** That is only the second rectangle. Add the first one, 12 times 8.
- **B:** That totals the side lengths rather than the areas. Multiply each rectangle's sides, then add the two products.
- **C:** That is only the first rectangle. Add the second one, 6 times 5.

Part B — correct answer: A

Why is decomposing a composite figure into rectangles a valid strategy?

A. The areas of the non-overlapping pieces add to the area of the whole figure.

B. Rectangles are the only shapes with area.

C. Any shape can be turned into one large rectangle.

D. The perimeter stays the same after decomposing.

Area is additive: if the pieces do not overlap and cover the figure completely, their areas sum to the total.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Counting a Region Twice

> Nia split a composite figure into a 10 by 4 rectangle and a triangle with base 10 and height 4. She added 40 + 20 = 60 square units — but the triangle sat ON TOP of part of the rectangle she had already counted.

Explain the error in Nia's method and state what she must check before adding areas.

Model answer: Decomposition only works when the pieces do not overlap. Nia counted the shared region twice, so 60 square units is too large. She must verify the pieces cover the figure exactly once before adding.

2 points	Student explains that decomposed pieces must not overlap, identifies that the overlapping region was counted twice, and states the total is therefore too large.
1 point	Student says the answer is wrong but does not explain the overlap.
0 points	Student agrees with Nia or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A T-shaped room can be decomposed in two different ways. Show both ways and prove they give the same total area. Use a T-shape where the top is 14 ft × 4 ft and the stem is 6 ft × 10 ft.

Model answer: Way 1 splits the T into its two named rectangles: 14 × 4 = 56 plus 6 × 10 = 60, a total of 116 sq ft. Way 2 wraps the T in a 14 ft × 14 ft rectangle worth 196 sq ft and subtracts the two empty 4 ft × 10 ft corners, 196 − 80 = 116 sq ft — the same region is being measured, so the decomposition you choose cannot change the area.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
